Thermodynamic Evaluation of the Cu-*R* (*R*: Ce, Pr, Nd, Sm) Binary Systems

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The phase diagrams and thermodynamic properties of the Cu-*R* (*R*: Ce, Pr, Nd, Sm) systems are reevaluated on the basis of all available experimental data, including the enthalpies of mixing reported recently. The Redlich-Kister polynomial with two or three terms is used to describe the excess Gibbs energies of liquid phases in the Cu-*R* systems. All intermediate phases are treated as stoichiometric, and their Gibbs energies of formation are optimized. Based on the optimized thermodynamic parameters, phase diagrams and thermodynamic properties of the Cu-*R* systems are calculated and compared with the experimental results. The optimized parameters seem to be in good agreement with the experimental data and might represent the regularity and systematic variation of these systems.

1. Introduction

There has been increasing interest by material scientists in rare earth (*R*) intermetallic compounds because of their potential physical and chemical characteristics. The great progress in rare earth superconductors has pushed forward research on the Cu-*R*-O ternary systems and their related multicomponent systems, in which Cu-*R* are the basic binary systems.

Table 1 Thermodynamic Properties of the Cu-Sm System

Lattice stability parameters for Cu

$^0G^{liq}_{Cu} = 0$

$^0G^{fcc}_{Cu} = -13\,054 + 9.613T$

Lattice stability parameters for Sm

$^0G^{liq}_{Sm} = 0$

$^0G^{bcc}_{Sm} = -8620 + 6.399T$

$^0G^{cph}_{Sm} = -11\,730 + 9.001T$

Integral molar Gibbs energies

$G^{liq} = X(1 - X)(-105\,784 + 62\,604X) + RT[X \ln X + (1 - X) \ln (1 - X)]$

$\Delta_f G\,(Cu_6Sm) = -28\,063 + 10.41T$

$\Delta_f G\,(Cu_5Sm) = -43\,175 + 21.99T$

$\Delta_f G\,(Cu_4Sm) = -31\,472 + 10.19T$

$\Delta_f G\,(Cu_2Sm) = -29\,651 + 4.22T$

$\Delta_f G\,(CuSm) = -39\,273 + 14.72T$

Note: From [88Sub2]. Gibbs energies are expressed in J/mol, and temperatures are in K. *X* is the mole fraction of Sm. Mol refers to the atom as the elementary entity.

The Cu-*R* systems were assessed and optimized by [88Sub1]. Since then, a number of new experimental data concerning the Cu-*R* systems were reported, and it is necessary to check and modify the optimized parameters by considering new experimental information. Furthermore, according to [91Oka] and [94Oka], the thermodynamic parameters of the Cu-*R* systems are open to some question. Here, the Cu-Sm system is used as an example.

Table 1 gives the thermodynamic parameters of the Cu-Sm system optimized by [88Sub2]. This table shows that the Gibbs energies of Cu_5Sm and CuSm are far more negative than those of other intermetallic compounds. That is to say, Cu_5Sm and CuSm would be extremely stable and other compounds in the system would become unstable at low temperature. Using the values given in Table 1, the Cu-Sm phase diagram was recalculated as shown in Fig. 1. As expected, the liquidus boundaries are almost the same as those shown by [88Sub2]. However, the compounds, except Cu_5Sm and CuSm, become unstable at low temperature, for which there is no evidence in the literature. Evidently, the thermodynamic modeling needs modification. Parameters of the Cu-Ce [88Sub3], Cu-Pr [88Sub4], and Cu-Nd [88Sub5] systems also share the same shortcoming. The Cu-Pr and Cu-Nd systems were evaluated by [92Wan] considering new experimental information. The thermodynamic properties of the liquid phases given by [92Wan] are in good agreement with experimental data. However, the thermodynamic parameters given by [92Wan] are open to the same question as [88Sub4] and [88Sub5], that is, at

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$T = 0$ K, the Gibbs energy of formation of Cu_4Nd is −357 kJ/mol, and that of CuNd is 81 kJ/mol.

In this work, all available experimental information of the Cu-R (R: Ce, Pr, Nd, Sm) systems is assessed and reevaluated, and a set of new self-consistent thermodynamic parameters are given.

2. Thermodynamic Models

2.1 Reference States

As reference states for these systems, the pure solid elements in their stable states at 298.15 K (stable element reference, SER) are chosen. The temperature dependence of the Gibbs energies of the elements were compiled by [91Din]. The equation appears in the form:

$$^0G^\phi(T) - H^{SER} = a + bT + cT\ln T + dT^2 + e/T + fT^3 + iT^4 + jT^7 + kT^{-9} \quad \text{(Eq 1)}$$

where in different temperature ranges different sets of coefficients a through k may be used.

2.2 Liquid Phases

The Gibbs energies of liquid phases can be divided into three parts: the reference part (ref), the ideal mixing part (id),and the excess part (ex).

$$G^{liq} - H^{SER} = {}^{ref}G^{liq} + {}^{id}G^{liq} + {}^{ex}G^{liq} \quad \text{(Eq 2)}$$

with

$$^{ref}G^{liq} = X_{Cu} \cdot [^0G^{liq}_{Cu}(T) - H^{SER}_{Cu}(298.15\text{ K})] + X_R \cdot [^0G^{liq}_R(T) - H^{SER}_R(298.15\text{ K})] \quad \text{(Eq 3)}$$

$$^{id}G^{liq} = RT(X_{Cu}\ln X_{Cu} + X_R \ln X_R) \quad \text{(Eq 4)}$$

The excess term is the function of the concentration represented by the Redlich-Kister polynomial [48Red]:

$$^{ex}G^{liq} = X_{Cu}X_R \sum_{i=0}^{n} {}^iL_{Cu,R}(X_{Cu} - X_R)^i \quad \text{(Eq 5)}$$

In the above equations, $^0G^\phi$ is the molar Gibbs energy of phase ϕ; X_{Cu} and X_R are mole fractions of Cu and R, respectively. H_i^{SER} (298.15 K) is the enthalpy of pure element i in its stable state at the reference temperature 298.15 K. H^{SER} is an abbreviation of $X_{Cu} \cdot H^{SER}_{Cu}(298.15\text{ K}) + X_R \cdot H^{SER}_R$ (298.15 K). $^0G^{liq}_{Cu}(T)$ and $^0G^{liq}_R(T)$ are Gibbs energies of pure liquid Cu and R at temperature T, respectively. $L_{Cu,R}$ represents the interaction parameters of liquid phases in the Cu-R systems that are to be optimized. The $L_{Cu,R}$ are functions of temperature described as linear in this evaluation.

2.3 Intermediate Phases

All intermediate phases are modeled as stoichiometric with Gibbs energies given by the formula:

$$^0G^\phi(T) - H^{SER} = X_{Cu} \cdot [^0G^{fcc}_{Cu}(T) - H^{SER}_{Cu}(298.15\text{ K})] + X_R \cdot [^0G^s_R(T) - H^{SER}_R(298.15\text{ K})] + a^\phi + b^\phi T \quad \text{(Eq 6)}$$

where s is the stable phase of R at 298.15 K, that is to say, s represents face-centered cubic (fcc) for Ce, double close-packed hexagonal (dcph) for Pr and Nd, and rhombohedral for Sm. a^ϕ and b^ϕ are parameters to be optimized. $(a^\phi + b^\phi T)$ can be regarded as the Gibbs energy of formation of phase ϕ.

3. Optimization and Calculation

The phase diagrams of all four systems were determined experimentally. Some thermodynamic properties, such as enthalpies of mixing of the liquid phases and heat contents of the liquid phases and intermediate phases, were measured too. All these data can be used to derive expressions for the thermodynamic functions of liquid phases and intermediate phases. In the present evaluation, the following assumptions are made.

(a) The melting temperatures, transformation temperatures, and thermodynamic functions of pure elements are from [91Din].

(b) Terminal solid solubilities in all four systems are not considered. Experiments do not indicate the existence of terminal solid solubility in the Cu-R systems, except that the maximum solid solubility of Ce in Cu was observed to be about 0.1 mol% Ce [64Dui, 71Kor] and the maximum solid solubilities of Cu in

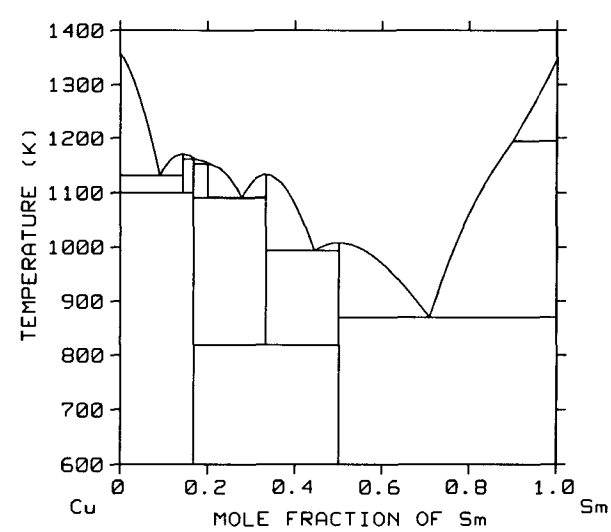

Fig. 1 Recalculated Cu-Sm phase diagram using the parameters given in Table 1.

Table 2 Thermodynamic Parameters of All Phases in the Cu-Ce System

Cu(fcc): ${}^0G^{fcc}_{Cu} - H^{SER}_{Cu} = \text{GHSERCU}$

Ce(fcc): ${}^0G^{fcc}_{Ce} - H^{SER}_{Ce} = \text{GHSERCE}$

Ce(bcc):

$${}^0G^{bcc}_{Ce} - H^{SER}_{Ce} = -1345.727 - 5.501344T - 7.6826482T\ln T - 0.029171596T^2 + 4.800907\text{E}-06T^3 - 196\,887T^{-1};\ 298.15 < T < 999.00;$$
$$-12\,101.271 + 187.446459T - 37.6142T\ln T;\ 999.00 < T < 1071.00$$
$$-11\,950.953 + 186.332816T - 37.4630853T\ln T - 5.7079\text{E}-05T^2 + 2.345\text{E}-09T^3$$
$$-25\,803T^{-1};\ 1071.00 < T < 4000.00$$

liquid:

$${}^0G^{liq}_{Cu} - H^{SER}_{Cu} = +5194.277 + 120.973331T - 24.112392T\ln T - 0.00265684T^2 + 1.29223E-07T^3 + 52\,478T^{-1} - 5.849E$$
$$-21T^7;\ 298.15 < T < 1357.77$$
$$-46.545 + 173.881484T - 31.38T\ln T;\ 1357.77 < T < 3200.00$$

$${}^0G^{liq}_{Ce} - H^{SER}_{Ce} = +4126.993 - 11.717588T - 7.4900290T\ln T - 0.02937917T^2$$
$$+4.844490\text{E}-06T^3 - 199\,424T^{-1};\ 298.15 < T < 999.00$$
$$-6730.688 + 183.015132T - 37.6978T\ln T;\ 999.00 < T < 2000.00$$

$${}^0L_{Cu,Ce} = -44\,377 - 4.03T$$
$${}^1L_{Cu,Ce} = -27\,205 + 2.93T$$
$${}^2L_{Cu,Ce} = -15\,465 + 9.88T$$

$$\text{GHSERCU} = -7770.458 + 130.485235T - 24.112392T\ln T - 0.00265684T^2$$

Cu_6Ce: ${}^0G^{Cu_6Ce} = 0.857{}^0G^{fcc}_{Cu} + 0.143{}^0G^{fcc}_{Ce} - 10\,113 - 1.08T$

Cu_5Ce: ${}^0G^{Cu_5Ce} = 0.833{}^0G^{fcc}_{Cu} + 0.167{}^0G^{fcc}_{Ce} - 11\,475 - 0.51T$

Cu_4Ce: ${}^0G^{Cu_4Ce} = 0.8{}^0G^{fcc}_{Cu} + 0.2{}^0G^{fcc}_{Ce} - 12\,411 - 0.56T$

Cu_2Ce: ${}^0G^{Cu_2Ce} = 0.667{}^0G^{fcc}_{Cu} + 0.333{}^0G^{fcc}_{Ce} - 14\,792 - 1.93T$

CuCe: ${}^0G^{CuCe} = 0.5{}^0G^{fcc}_{Cu} + 0.5{}^0G^{fcc}_{Ce} - 12\,708 - 0.59T$

$$+1.29223\text{E}-07T^3 + 52\,478T^{-1};\ 298.15 < T < 1357.77$$
$$-13\,542.026 + 183.803828T - 31.38T\ln T + 3.64167\text{E}+29T^{-9};\ 1357.77 < T < 3200.00$$

$$\text{GHSERCE} = -7160.520 + 84.229821T - 22.3664T\ln T - 0.0067103T^2$$
$$-3.20773\text{E}-07T^3 - 18\,117T^{-1};\ 298.15 < T < 999.00$$
$$-79\,235.849 + 656.248996T - 100.886328T\ln T + 0.0258680 39T^2$$
$$-1.917077\text{E}-06T^3 + 11\,452\,449T^{-1};\ 999.00 < T < 2000.00$$
$$-14\,205.264 + 190.374747T - 37.6978T\ln T;\ 2000.00 < T < 4000.00$$

Note: Gibbs energies are expressed in J/mol, and temperatures are in K. Mol refers to the atom as the elementary entity. Parameters of pure Cu and Ce are from [91Din].

fcc (Ce) and body-centered cubic (bcc) (Ce) were derived to be about 0.55 and 0.37 mol% Cu, respectively [61Gsc, 88Sub3].

(c) All intermediate phases are modeled as stoichiometric compounds, as described above.

(d) According to [79Kub] and [93Tan], a fixed ratio exists between excess entropy and enthalpy. In the modeling of excess Gibbs energies of the liquid phases and Gibbs energies of formation of the intermediate phases, a constraint between entropy and enthalpy is imposed because the experimental data are not sufficient to optimize the entropy and enthalpy independently. An attempt to optimize entropy and enthalpy independently for intermediate phases may result in unreasonable decomposition of the solid phases at room temperature, for which there is no experimental evidence.

The optimization of all four binary systems was performed using the BINGSS program developed by Lukas et al. [92Luk]. The final calculation of the binary phase diagrams and thermodynamic properties was performed by using the Thermo-Calc program package developed by Sundman et al. [85Sun].

3.1 Cu-Ce System

The Cu-Ce phase diagram was first determined by thermal analysis by [15Han], who did not report the existence of Cu_5Ce. Another phase diagram [61Gsc] also did not indicate the formation of Cu_5Ce. This phase was first reported by [61Dwi]. Subsequently, [64Rhi] confirmed the existence of Cu_5Ce from electron microprobe x-ray analysis of metallographic samples and from single-crystal analysis. The existence of Cu_5Ce in the Cu-Ce system was accepted in a subsequent revision [74Gsc]. However, although a disagreement appeared on the existence of Cu_5Ce, the Cu-Ce liquidus boundaries are in good agreement with each other [15Han, 64Rhi, 74Gsc].

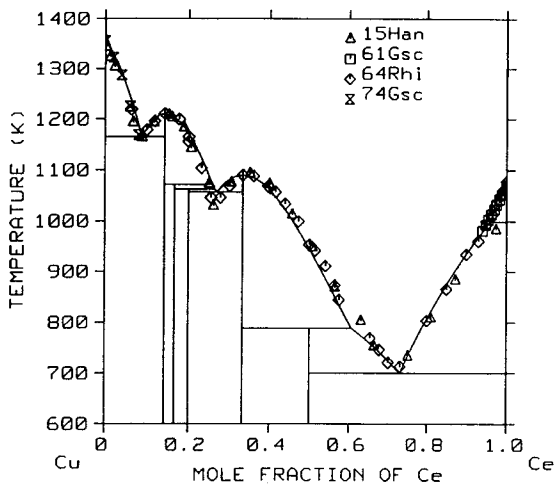

Fig. 2 Calculated Cu-Ce phase diagram.

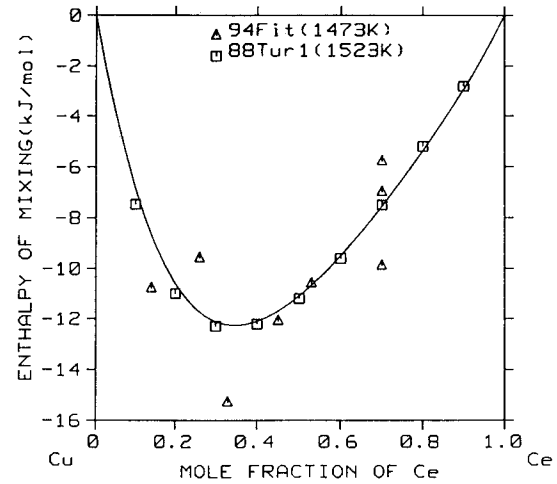

Fig. 3 Calculated enthalpies of mixing of the liquid Cu-Ce alloys at 1500 K.

The partial enthalpies of Cu and Ce in their binary alloys at 1523 K were measured by a high-temperature calorimeter, and the enthalpies of mixing were obtained by adding the relevant partial terms [88Tur1]. The enthalpies of mixing of liquid Cu-Ce alloys were also measured by high-temperature calorimetry at 1473 K using untreated commercial copper powder [93Fit]. The discrepancy between these two results [88Tur1, 93Fit] appeared to be very large. So, Fitzner and Kleppa remeasured the enthalpies of mixing of liquid Cu-Ce alloys using either small pieces of copper wire or copper powder sintered into small pieces in a stream of hydrogen gas at about 873 K [94Fit], which are much less negative than those of [93Fit], but in good agreement with those of [88Tur1].

The heat contents of several Cu-Ce alloys with different compositions were determined over the temperature range 1400 to 2300 K using a levitation calorimeter [80Dok] and over the temperature range 850 to 1300 K using a drop calorimeter [89Qi].

Experimental phase diagram data [15Han, 61Gsc, 64Rhi, 74Gsc], enthalpies of mixing [94Fit, 88Tur1], and heat contents [80Dok, 89Qi] are used in the optimization of the Cu-Ce system. Table 2 gives the optimized parameters of all phases in the Cu-Ce system. The phase diagrams and thermodynamic properties of the Cu-Ce system were calculated using these parameters. The calculated Cu-Ce phase diagram is shown in Fig. 2. The agreement between the calculated and experimental phase diagram data is good. Table 3 presents the calculated special points in the Cu-Ce phase diagram. Figure 3 shows the calculated enthalpies of mixing of liquid in the Cu-Ce system at 1500 K with the experimental data [94Fit, 88Tur1]. Inspection of Fig. 3 indicates that the calculated values agree well with the experimental data. The calculated heat contents of two alloys with $X_{Ce} = 0.143$ (Fig. 4) and $X_{Ce} = 0.1995$ (Fig. 5) are compared with the experimental data [80Dok, 89Qi]. It is demonstrated that the experimental data are well described by the present calculation within the experimental errors.

3.2 Cu-Pr System

The Cu-Pr phase diagram without Cu_5Pr was determined early [34Can]. The existence of Cu_5Pr with the hexagonal $CaCu_5$ prototype structure was identified based on x-ray diffraction (XRD) analysis [61Dwi] and was accepted in the assessed Cu-Pr phase diagram [88Sub4]. According to the formation rule of the other Cu_5R phases, Cu_5Pr is likely to form from Cu_6Pr and liquid through a peritectic reaction at about 1100 K, which is determined by interpolation of the melting data of other Cu_5R phases. The congruent melting temperatures of the compounds Cu_6Pr and Cu_2Pr in the phase diagram [34Can] are about 10 K higher than the values obtained by interpolation of the data for the other Cu-*R* systems.

The enthalpies of mixing of liquid Cu-Pr alloys were measured by high-temperature calorimetry at 1473 K [94Fit] and at 1523 K [88Tur2]. Unfortunately, a significant difference (about 4 kJ/mol in the middle of the system) between them exists. According to [94Fit], this difference may be due to several factors, such as the small difference in experimental temperatures (1473 versus 1523 K); different materials used for crucibles (BN and BeO in [94Fit], Al_2O_3 and Mo in [88Tur2]); purity of the rare earth; and accepted values of the enthalpies of the pure metals. The heat contents of six Cu-Pr binary alloys with the respective compositions $X_{Pr} = 0.087$, 0.143, 0.169, 0.294,

Table 3 Calculated Special Points in the Cu-Ce Phase Diagram

Reaction	Temperature, K	Composition of liquid (X_{Ce})	Reaction type
liq ↔ Cu(fcc) + Cu_6Ce	1165	0.088	Eutectic
liq ↔ Cu_6Ce	1211	0.143	Congruent
liq + Cu_6Ce ↔ Cu_5Ce	1072	0.257	Peritectic
liq + Cu_5Ce ↔ Cu_4Ce	1063	0.263	Peritectic
liq ↔ Cu_4Ce + Cu_2Ce	1057	0.268	Eutectic
liq ↔ Cu_2Ce	1090	0.333	Congruent
liq + Cu_2Ce ↔ CuCe	788	0.606	Peritectic
liq ↔ CuCe + Ce(fcc)	700	0.730	Eutectic
Ce(bcc) ↔ liq + Ce(fcc)	999	0.958	Catatectic

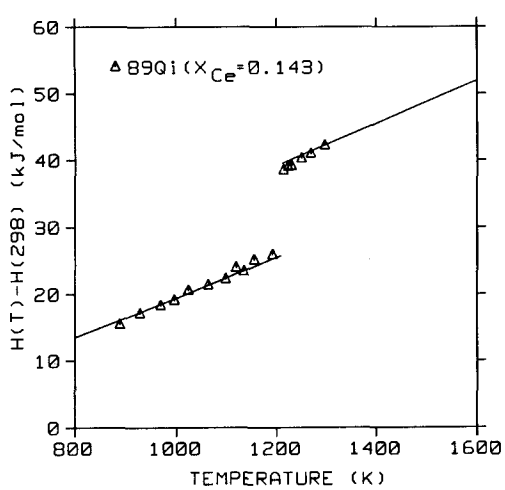

Fig. 4 Comparison between calculated and measured heat contents of a Cu-Ce alloy with X_{Ce} = 0.143.

0.333, and 0.500, over the temperature range 850 to 1300 K were measured using a drop calorimeter [89Qi].

Phase diagram data [34Can], enthalpies of mixing [94Fit], and heat contents [89Qi] were used in the optimization of the Cu-Pr system. The congruent melting temperatures of the compounds Cu_6Pr and Cu_2Pr are from [89Qi] instead of [34Can]. Table 4 gives optimized parameters of all phases in the Cu-Pr system. The phase diagram and thermodynamic properties of the Cu-Pr system were calculated using the parameters given in Table 4. Figure 6 shows the calculated Cu-Pr phase diagram with the measured data [34Can]. Table 5 presents the calculated special points in the Cu-Pr phase diagram. In view of the experimental errors, the calculated phase diagram is satisfactory. Figure 7 shows the calculated enthalpies of mixing of the liquid Cu-Pr alloys at 1500 K with the measured data [94Fit, 88Tur2]. The calculated values agree with those measured by [94Fit]. The comparison between calculated and measured heat contents of a Cu-Pr alloy (X_{Pr} = 0.087) is presented in Fig. 8. Within the experimental errors, the agreement is excellent.

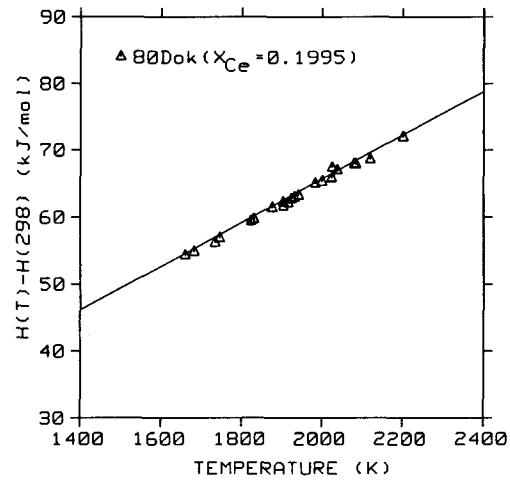

Fig. 5 Comparison between calculated and measured heat contents of a Cu-Ce alloy with X_{Ce} = 0.1995.

3.3 Cu-Nd System

[83Car] and [83Zhe] independently determined the Cu-Nd phase diagram using differential thermal analysis (DTA), metallography, and XRD analysis. Later, [84Lak] examined the Cu-rich side of this system using scanning electron microscopy in addition to the above techniques. [88Efi] also constructed the Cu-Nd phase diagram in the composition range 0 to 20 mol% Nd, but only two liquidus data points were given by DTA results. [90Gu] investigated the entire system by means of DTA and XRD analysis. [83Car], [83Zhe], and [90Gu] all reported the existence of Cu_6Nd, Cu_5Nd, Cu_4Nd, Cu_2ND, and CuNd, whereas [88Efi] did not report the existence of Cu_6Nd. [83Car] reported the high-temperature formation of Cu_7Nd_2 at 1098 K with its subsequent decomposition at 1058 K whereas this phase is reported to be retained at room temperature by [90Gu]. However, the presence of Cu_7Nd_2 is not corroborated by either [83Zhe] or [84Lak]. The discrepancy among the DTA results of all the reports also appears to be very large. For example, as to the temperature of the peritectic reaction liq + Cu_2Nd ↔ CuNd, the difference between the results of [83Zhe] (875 K) and [84Lak] (964 K) is 89 K. The congruent melting temperature of Cu_6Nd reported by [83Car] is 1183 K, whereas 1235 K is reported by [83Zhe].

Table 4 Thermodynamic Parameters of All Phases in the Cu-Pr System

Cu (fcc): ${}^0G_{Cu}^{fcc} - H_{Cu}^{SER}$ = GHSERCU

Pr(dcph): ${}^0G_{Pr}^{dcph} - H_{Pr}^{SER}$ = GHSERPR

Pr (bcc):

${}^0G_{Pr}^{bcc} - H_{Pr}^{SER} =$

$-2863.651 + 28.274853T - 13.7470527T \ln T - 0.02284377T^2$

$+3.542468E-06T^3 - 87\,486T^{-1}$; $298.15 < T < 1068.00$

$-11\,985.919 + 188.657121T - 38.451T \ln T$; $1068.00 < T < 1204.00$

$+953.224 + 100.826281T - 26.6824313T \ln T - 0.004106833T^2$

$+1.76214E-07T^3 - 2\,473\,024T^{-1}$; $1204.00 < T < 3800.00$

liquid:

${}^0G_{Cu}^{liq} - H_{Cu}^{SER}$ = (the same as in Table 2)

${}^0G_{Pr}^{liq} - H_{Pr}^{SER} = +3848.961 - 29.099465T - 4.7344931T \ln T - 0.035119723T^2$

$+5.427467E-06T^3 - 207\,406T^{-1}$; $298.15 < T < 1068.00$

$-10\,539.574 + 219.508805T - 42.9697T \ln T$; $1068.00 < T < 3800.00$

${}^0L_{Cu,Pr} = -67\,977 + 6.80T$

${}^1L_{Cu,Pr} = -29\,602 + 2.96T$

Cu_6Pr: ${}^0G^{Cu_6Pr} = 0.857{}^0G_{Cu}^{fcc} + 0.143{}^0G_{Pr}^{dcph} - 12\,708 + 0.05T$

Cu_5Pr: ${}^0G^{Cu_5Pr} = 0.833{}^0G_{Cu}^{fcc} + 0.167{}^0G_{Pr}^{dcph} - 13\,755 + 0.11T$

Cu_4Pr: ${}^0G^{Cu_4Pr} = 0.8{}^0G_{Cu}^{fcc} + 0.2{}^0G_{Pr}^{dcph} - 15\,059 + 0.12T$

Cu_2Pr: ${}^0G^{Cu_2Pr} = 0.667{}^0G_{Cu}^{fcc} + 0.333{}^0G_{Pr}^{dcph} - 19\,687 + 0.21T$

CuPr: ${}^0G^{CuPr} = 0.5{}^0G_{Cu}^{fcc} + 0.5{}^0G_{Pr}^{dcph} - 19\,046 + 2.50T$

GHSERCU = (the same as in Table 2)

GHSERPR $= -18\,803.379 + 356.587384T - 68.9176T\ln T + 0.072929T^2$
$- 2.5184333E-05T^3 + 507\,385T^{-1}$; $298.15 < T < 500.00$
$- 7246.848 + 82.427384T - 22.8909T\ln T - 0.00497126T^2$
$- 1.22951E-06T^3$; $500.00 < T < 800.00$
$+ 95\,411.023 - 1073.55111T + 146.764T\ln T - 0.1288205T^2$
$+ 1.5592233E-05T^3 - 11\,588\,800T^{-1}$; $800.00 < T < 1068.00$
$- 481\,663.131 + 4234.33311T - 606.120311T\ln T + 0.305181506T^2$
$- 3.0994702E-05T^3 + 70\,926\,840T^{-1}$; $1068.00 < T < 1204.00$
$- 20\,014.678 + 227.685155T - 42.9697T\ln T$; $1204.00 < T < 3800.00$

Note: Gibbs energies are expressed in J/mol, and temperatures are in K. Mol refers to the atom as the elementary entity. Parameters of pure Cu and Pr are from [91Din].

The enthalpies of mixing of liquid Cu-Nd alloys were measured by high-temperature calorimetry at 1473 K [94Fit] and at 1523 K [88Tur2]. The discrepancy between the results is very large. The reason for this discrepancy was discussed in [94Fit]. The heat contents of several Cu-Nd binary alloys above 850 K were measured using a drop calorimeter [89Qi].

Experimental phase diagram data [83Car, 83Zhe, 84Lak, 88Efi, 90Gu], enthalpies of mixing [94Fit], and heat contents [89Qi] were used in the optimization of the Cu-Nd system. The optimized parameters of all phases in the Cu-Nd system are presented in Table 6. The phase diagram and thermodynamic properties of the Cu-Nd system were calculated using the pa-

Table 5 Calculated Special Points in the Cu-Pr Phase Diagram

Reaction	Temperature, K	Composition of liquid (X_{Pr})	Reaction type
liq ↔ Cu(fcc) + Cu_6Pr	1169	0.086	Eutectic
liq ↔ Cu_6Pr	1218	0.143	Congruent
liq + Cu_6Pr ↔ Cu_5Pr	1104	0.243	Peritectic
liq + Cu_5Pr ↔ Cu_4Pr	1093	0.249	Peritectic
liq ↔ Cu_4Pr + Cu_2Pr	1069	0.270	Eutectic
liq ↔ Cu_2Pr	1104	0.333	Congruent
liq + Cu_2Pr ↔ CuPr	855	0.530	Peritectic
liq ↔ CuPr + Pr(dcph)	715	0.698	Eutectic
Pr(bcc) ↔ liq + Pr(dcph)	1068	0.912	Catatectic

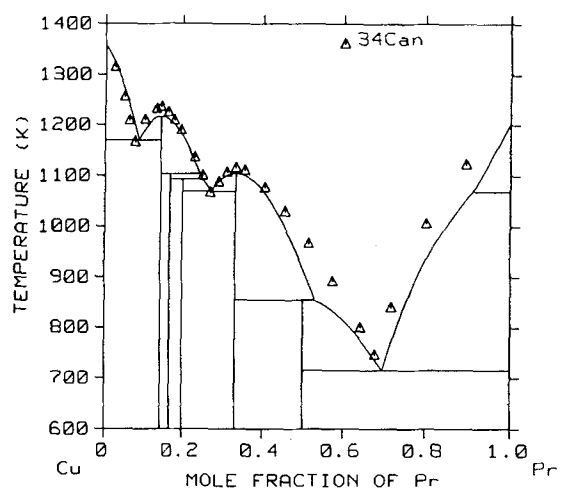

Fig. 6 Calculated Cu-Pr phase diagram.

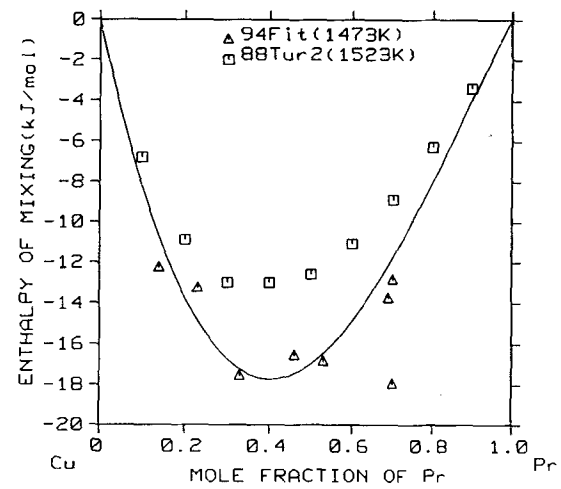

Fig. 7 Calculated enthalpies of mixing of the liquid Cu-Pr alloys at 1500 K.

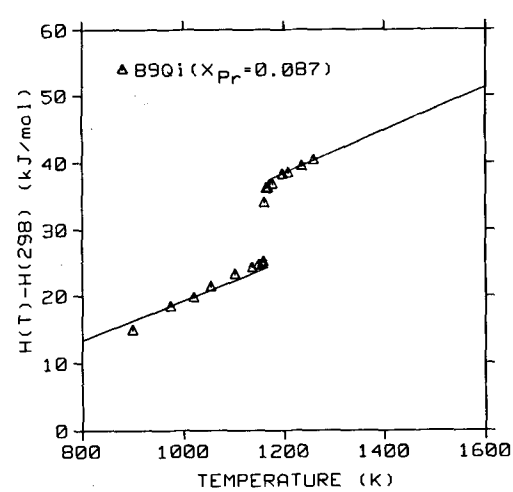

Fig. 8 Comparison between calculated and measured heat contents of a Cu-Pr alloy with $X_{Pr} = 0.087$.

rameters presented in Table 6. Figure 9 shows the calculated Cu-Nd phase diagram with the measured data [83Car, 83Zhe, 84Lak, 88Efi, 90Gu]. Table 7 presents the calculated special points in the Cu-Nd phase diagram. In view of the experimental scatter band, most of the experimental values are reasonably reproduced by the calculation. Figure 10 shows the calculated enthalpies of mixing of the liquid Cu-Nd alloys at 1500 K with measured data [94Fit, 88Tur2]. The calculated values agree with those measured by [94Fit]. The comparison between calculated and measured heat contents of a Cu-Nd alloy ($X_{Nd} = 0.116$) is presented in Fig. 11. The agreement is excellent within the experimental errors.

After the present manuscript was finished, an assessment of the Cu-Nd system by [96Du] was published. The calculated phase diagram and thermodynamic properties of this system agree with the present work. However, [96Du] estimated the

Table 6 Thermodynamic Parameters of All Phases in the Cu-Nd System

Cu (fcc): ${}^0G^{fcc}_{Cu} - H^{SER}_{Cu}$ = GHSERCU

Nd (dcph): ${}^0G^{dcph}_{Nd} - H^{SER}_{Nd}$ = GHSERND

Nd(bcc):

${}^0G^{bcc}_{Nd} - H^{SER}_{Nd} =$

$+ 1437.295 - 0.546281T$ + GHSERND; $298.15 < T < 400.00$

$+ 7312.2 - 153.033976T + 14.9956777T \ln T - 0.050479T^2$

$+ 7.287217E-06T^3 - 831\,810T^{-1}$; $400.00 < T < 1128.00$

$- 18\,030.266 + 239.677322T - 44.5596T \ln T$; $1128.00 < T < 1289.00$

$+ 334\,513.017 - 2363.9199T + 311.409193T \ln T - 0.156030778T^2$

$+ 1.2408421E-05T^3 - 64\,319\,604T^{-1}$; $1289.00 < T < 1800.00$

liquid:

${}^0G^{liq}_{Cu} - H^{SER}_{Cu}$ = (the same as in Table 2)

${}^0G^{liq}_{Nd} - H^{SER}_{Nd} = + 5051.743 - 1.585076T$ + GHSERND; $298.15 < T < 300.00$

$+ 5350.01 - 86.593963T + 5.357301T \ln T - 0.046955463T^2$

$+ 6.860782E-06T^3 - 374\,380T^{-1}$; $300.00 < T < 1128.00$

$- 16\,335.232 + 268.625903T - 48.7854T \ln T$; $1128.00 < T < 1800.00$

${}^0L_{Cu,Nd} = -71\,130 + 7.11T$

${}^1L_{Cu,Nd} = -31\,414 + 3.14T$

Cu_6Nd: ${}^0G^{Cu_6Nd} = 0.857{}^0G^{fcc}_{Cu} + 0.143{}^0G^{dcph}_{Nd} - 13\,301 + 0.17T$

Cu_5Nd: ${}^0G^{Cu_5Nd} = 0.833{}^0G^{fcc}_{Cu} + 0.167{}^0G^{dcph}_{Nd} - 14\,375 + 0.20T$

Cu_4Nd: ${}^0G^{Cu_4Nd} = 0.8{}^0G^{fcc}_{Cu} + 0.2{}^0G^{dcph}_{Nd} - 15\,718 + 0.21T$

Cu_2Nd: ${}^0G^{Cu_2Nd} = 0.667{}^0G^{fcc}_{Cu} + 0.333{}^0G^{dcph}_{Nd} - 20\,498 + 0.37T$

CuNd: ${}^0G^{CuNd} = 0.5{}^0G^{fcc}_{Cu} + 0.5{}^0G^{dcph}_{Nd} - 20\,047 + 2.51T$

GHSERCU = (the same as in Table 2)

GHSERND $= -8402.93 + 111.10239T - 27.0858T \ln T + 5.56125E-04T^2$

$- 2.6923E-06T^3 + 34\,887T^{-1}$; $298.15 < T < 900.00$

$- 6984.083 + 83.662617T - 22.7536T \ln T - 0.00420402T^2$

$-1.802E-06T^3$; $900.00 < T < 1128.00$

$- 225\,610.846 + 1673.04075T - 238.182873T \ln T + 0.078615997T^2$

$- 6.048207E-06T^3 + 38\,810\,350T^{-1}$; $1128.00 < T < 1799.00$

$- 25\,742.331 + 276.257088T - 48.7854T \ln T$; $1799.00 < T < 1800.00$

Note: Gibbs energies are expressed in J/mol, and temperatures are in K. Mol refers to the atom as the elementary entity. Parameters of pure Cu and Nd are from [91Din].

Gibbs energy of formation of Cu_7Nd_2. In the calculation of [96Du], Cu_7Nd_2 only exists within a very limited temperature range, 1070 to 1094 K.

3.4 Cu-Sm System

The Cu-Sm phase diagram was determined by several investigators [64Cop, 73Tor, 75Kuh, 77Gol]. The investigation of [64Cop] was partial and restricted to the region between 50 and 100 mol% Sm. All other investigated results are in agreement with regard to the existence of Cu_6Sm, Cu_2Sm, and CuSm. However, [73Tor] reported a phase diagram without Cu_4Sm, whereas the diagram of [77Gol] does not show Cu_5Sm. All five intermediate phases mentioned above appeared in the phase diagram of [75Kuh], which is based on

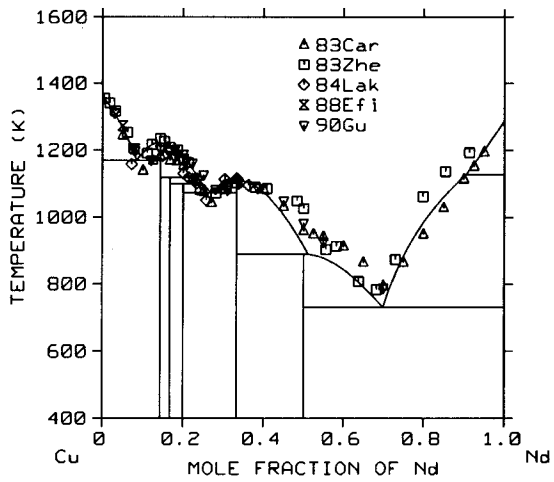

Fig. 9 Calculated Cu-Nd phase diagram.

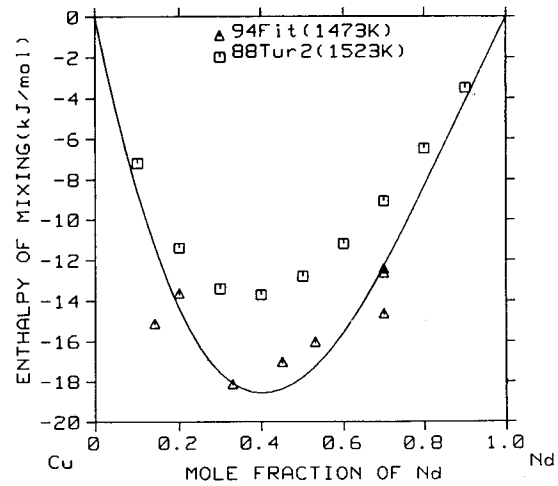

Fig. 10 Calculated enthalpies of mixing of the liquid Cu-Nd alloys at 1500 K.

Table 7 Calculated Special Points in the Cu-Nd Phase Diagram

Reaction	Temperature, K	Composition of liquid (X_{Nd})	Reaction type
liq ↔ Cu(fcc) + Cu_6Nd	1169	0.084	Eutectic
liq ↔ Cu_6Nd	1223	0.143	Congruent
liq + Cu_6Nd ↔ Cu_5Nd	1118	0.236	Peritectic
liq + Cu_5Nd ↔ Cu_4Nd	1100	0.247	Peritectic
liq ↔ Cu_4Nd + Cu_2Nd	1073	0.269	Eutectic
liq ↔ Cu_2Nd	1111	0.333	Congruent
liq + Cu_2Nd ↔ CuNd	889	0.511	Peritectic
liq ↔ CuNd + Nd(dcph)	732	0.699	Eutectic
Nd(bcc) ↔ liq + Nd(dcph)	1128	0.905	Catatectic

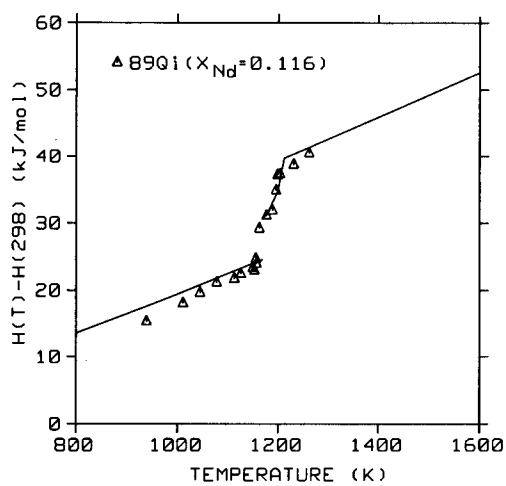

Fig. 11 Comparison between calculated and measured heat contents of a Cu-Nd alloy with X_{Nd} = 0.116.

DTA, metallography, and electron microprobe analysis. But the congruent melting temperature of Cu_6Sm in [75Kuh] is lower by about 30 K than that measured by other investigators [73Tor, 74Rus, 77Gol]. The diagram measured by [73Tor] showed that CuSm was a congruently melting compound with melting temperature at 1063 K, whereas CuSm was reported by [64Cop], [74Rus], [75Kuh], and [77Gol] to decompose peritectically. The reported peritectic temperatures disagree with each other (from 993 to 1103 K). On the other hand, the congruent melting of the equiatomic Cu*R* phase was observed for the heavy rare earth metals (Gd, Dy, and Er). In the evalu-

ation of [88Sub2], CuSm was considered a congruently melting compound.

Because of considerable conflicts among the reported phase relationships and melting temperatures, the accepted data are those which agree with the systematics of the Cu-*R* systems. All five intermediate phases are accepted in this work, and CuSm is regarded as a congruently melting compound. The liquidus temperatures between 40 and 70 mol% Sm reported in [64Cop] are not accepted because these temperatures are much higher than those of other investigators.

The enthalpies of mixing of the liquid Cu-Sm alloys were measured by high-temperature calorimetry at 1390 K [88Tur3]. According to [94Fit] and the general trend of the enthalpies of mixing of liquid Cu-*R* alloys, the measured data of [88Tur3] may be less negative. No other experimental thermodynamic data are available for this system.

In the present evaluation, the experimental Cu-Sm phase diagram data were utilized to derive analytical expressions for the Gibbs energy function of the liquid phase and the Gibbs energies of formation of the intermediate phases. The optimized parameters of all phases in the Cu-Sm system are presented in Table 8. Figure 12 shows the calculated Cu-Sm phase diagram with the measured data [64Cop, 73Tor, 75Kuh, 77Gol]. Table 9 presents the calculated special points in the Cu-Sm phase

Table 8 Thermodynamic Parameters of All Phases in the Cu-Sm System

Cu (fcc): ${}^0G^{fcc}_{Cu} - H^{SER}_{Cu} = \text{GHSERCU}$

Sm (rhomb): ${}^0G^{rhomb}_{Sm} - H^{SER}_{Sm} = \text{GHSERSM}$

Sm (bcc):

$$
\begin{aligned}
{}^0G^{bcc}_{Sm} - H^{SER}_{Sm} = & -4368.72 + 55.972523T - 16.9298494T\ln T - 0.025446016T^2 \\
& + 3.579527\text{E}-06T^3 + 94\,209T^{-1};\ 298.15 < T < 1190.00 \\
& - 15\,957.862 + 253.121044T - 46.9445T\ln T;\ 1190.00 < T < 1345.00 \\
& + 111\,191.653 - 624.680805T + 71.6856914T\ln T - 0.047314968T^2 \\
& + 3.329865\text{E}-06T^3 - 24\,870\,276T^{-1};\ 1345.00 < T < 2100.00
\end{aligned}
$$

liquid:

${}^0G^{liq}_{Cu} - H^{SER}_{Cu} =$ (the same as in Table 2)

$$
\begin{aligned}
{}^0G^{liq}_{Sm} - H^{SER}_{Sm} = & + 3468.783 + 20.117456T - 11.6968284T\ln T - 0.03241817\,T^2 \\
& + 4.544272\text{E}-06T^3 + 23\,528T^{-1};\ 298.15 < T < 1190.00 \\
& - 11\,728.229 + 273.487076T - 50.208T\ln T;\ 1190.00 < T < 2100.0
\end{aligned}
$$

${}^0L_{Cu,Sm} = -75\,873 + 7.59T$

${}^1L_{Cu,Sm} = -28\,692 + 2.87T$

Cu_6Sm: ${}^0G^{Cu_6Sm} = 0.857{}^0G^{fcc}_{Cu} + 0.143{}^0G^{rhomb}_{Sm} - 13\,647 + 0.11T$

Cu_5Sm: ${}^0G^{Cu_5Sm} = 0.833{}^0G^{fcc}_{Cu} + 0.167{}^0G^{rhomb}_{Sm} - 15\,026 + 0.16T$

Cu_4Sm: ${}^0G^{Cu_4Sm} = 0.8{}^0G^{fcc}_{Cu} + 0.2{}^0G^{rhomb}_{Sm} - 16\,598 + 0.24T$

Cu_2Sm: ${}^0G^{Cu_2Sm} = 0.667{}^0G^{fcc}_{Cu} + 0.333{}^0G^{rhomb}_{Sm} - 21\,131 + 0.26T$

CuSm: ${}^0G^{CuSm} = 0.5{}^0G^{fcc}_{Cu} + 0.5{}^0G^{rhomb}_{Sm} - 20\,855 + 0.27T$

GHSERCU = (the same as in Table 2)

$$
\begin{aligned}
\text{GHSERSM} = & -3872.013 - 32.10748T - 1.6485T\ln T - 0.050254T^2 \\
& + 1.010345\text{E}-05T^3 + 82\,168T^{-1};\ 298.15 < T < 700.00 \\
& - 50\,078.215 + 627.869894T - 102.665T\ln T + 0.0474522T^2 \\
& -7.538383\text{E}-06T^3 + 3\,861\,770T^{-1};\ 700.00 < T < 1190.00 \\
& + 289\,719.819 - 2744.50976T + 381.41982T\ln T - 0.254986338T^2 \\
& + 2.7512152\text{E}-05T^3 - 40\,102\,102T^{-1};\ 1190.00 < T < 1345.00 \\
& - 23\,056.079 + 282.194375T - 50.208T\ln T;\ 1345.00 < T < 2100.00
\end{aligned}
$$

Note: Gibbs energies are expressed in J/mol, and temperatures are in K. Mol refers to the atom as the elementary entity. Parameters of pure Cu and Sm are from [91Din].

diagram. In view of the experimental scatter band, the fit to the experimental values is satisfactory. Figure 13 shows the calculated enthalpies of mixing of liquid Cu-Sm alloys at 1500 K with the measured data [88Tur3]. The calculated values are more negative than those measured by [88Tur3].

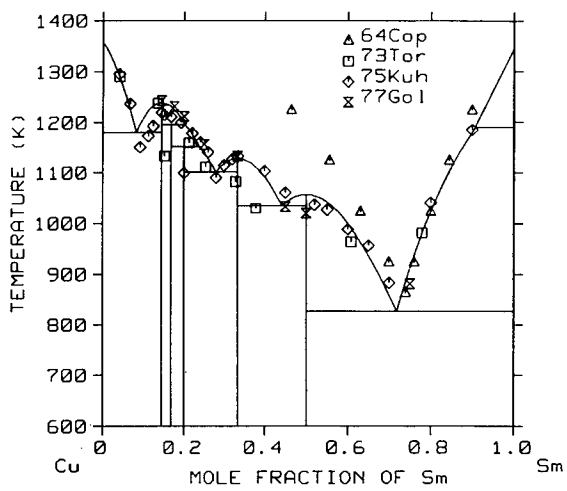

Fig. 12 Calculated Cu-Sm phase diagram.

3.5 Variation of the Cu-*R* Phase Diagrams and Thermodynamic Properties

From the results of the evaluation and the experimental results for these four systems, some regularity and systematic variation can be observed. In the following, the variation of melting temperatures of the intermediate phases, eutectic temperatures of the systems, excess thermodynamic properties of the liquid phases, and the enthalpies of formation of the intermediate phases are discussed.

3.5.1 Melting Temperatures of Intermediate Phases

In all investigated systems, five intermediate phases were observed with the following stoichiometries: Cu_6R, Cu_5R, Cu_4R, Cu_2R, and CuR. Of these intermediate phases, Cu_6R and Cu_2R melt congruently, and the others, except CuSm, melt incongruently. Figure 14 presents the variation of the calculated melting temperatures, which increase with decrease of atomic radius of *R*. The incongruently melting temperatures increase more quickly than the congruently melting temperatures.

3.5.2 Eutectic Temperatures

Three eutectic points are observed in the Cu-*R* systems, except the Cu-Sm system, in which four eutectics are observed. These eutectics are located in the following two-phase regions: Cu-Cu_6R, Cu_4R-Cu_2R, and Cu-*R*-*R*. Figure 15 presents the variation of the calculated eutectic temperatures in the Cu-*R* systems. Inspection of Fig. 15 indicates that the eutectic tem-

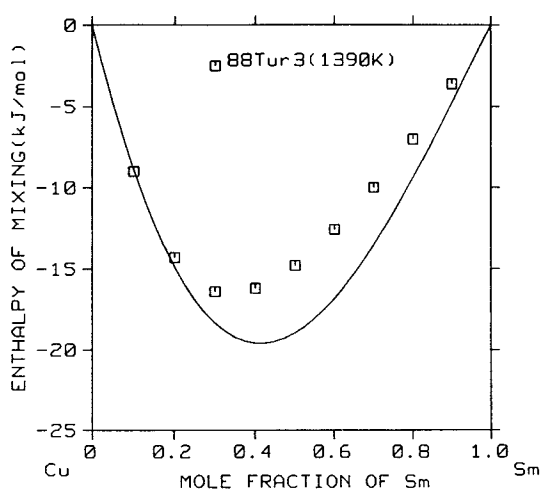

Fig. 13 Calculated enthalpies of mixing of the liquid Cu-Sm alloys at 1500 K.

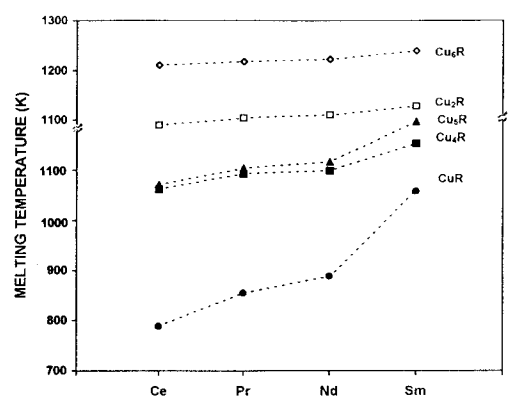

Fig. 14 Melting temperatures of the Cu-*R* intermediate phases.

peratures increase with decrease of atomic radius of R, where the temperatures of the Cu-Cu_6R and Cu_4R-Cu_2R eutectics increase slowly and the temperatures of the CuR-R eutectics increase quickly.

3.5.3 Excess Thermodynamic Properties of Liquid Phases

The enthalpies of mixing and excess Gibbs energies of liquid phases in the Cu-R systems are all negative and exhibit deep minima in their compositional dependence with minimal values all located at $X_R = 0.4$ nearby. Figure 16 shows the variation of the minimal values of the calculated enthalpies of mixing and excess Gibbs energies of the liquid phases in the Cu-R systems at 1500 K. The negative values of the enthalpies of mixing and excess Gibbs energies of liquid phases increase with decrease of atomic radius of R.

3.5.4 Enthalpies of Formation of Intermediate Phases

Table 10 presents the enthalpies of formation of intermediate phases in the Cu-R systems obtained in different ways. The reference states in [88Sub2], [88Sub3], [88Sub4], and [88Sub5] are liquid Cu and liquid R. These results, therefore, cannot be compared with the others. However, no systematic changes can be observed from these results. The values optimized in this evaluation are most close to the measured data [94Fit]. The values derived on the basis of the semiempirical model of Miedema [80Mie, 83Nie] are all more negative than those optimized in this evaluation and measured by [94Fit]. According to [94Fit], while the Miedema model correctly predicts overall trends in the enthalpies of formation, the model generally overestimates the negative values.

Figure 17 presents the optimized enthalpies of formation of intermediate phases. The enthalpies of formation exhibit V-type compositional dependence where the minimal values are observed at Cu_2R. The negative values of the enthalpies of formation of phases with the same stoichiometry increase with decrease of atomic radius of R.

4. Conclusions

Experimental thermodynamic and phase equilibrium data of the Cu-R (R: Ce, Pr, Nd, Sm) systems are assessed and optimized, and a set of self-consistent thermodynamic parameters are now available. The optimized thermodynamic parameters seem to be in good agreement with the experimental data, and these parameters might represent the regularity and systematic variation of these systems. However, some problems still remain to be resolved. (a) The terminal solid solubility in the Cu-Ce system should be determined. (b) The Cu-Pr phase diagram should be experimentally determined more carefully, especially with regard to the liquidus boundaries. (c) The existence and the stable temperature range of Cu_7Nd_2 should be checked more carefully. (d) The congruent melting of CuSm should be checked with care. (e) More thermodynamic data should be measured, especially the excess entropies of liquid phases.

Table 9 Calculated Special Points in the Cu-Sm Phase Diagram

Reaction	Temperature, K	Composition of liquid (X_{Sm})	Reaction type
liq ↔ Cu(fcc) + Cu_6Sm	1180	0.081	Eutectic
liq ↔ Cu_6Sm	1240	0.143	Congruent
liq + Cu_6Sm ↔ Cu_5Sm	1196	0.243	Peritectic
liq + Cu_5Sm ↔ Cu_4Sm	1153	0.249	Peritectic
liq ↔ Cu_4Sm + Cu_2Sm	1102	0.270	Eutectic
liq ↔ Cu_2Sm	1128	0.333	Congruent
liq ↔ Cu_2Sm + CuSm	1035	0.443	Eutectic
liq ↔ CuSm	1058	0.500	Congruent
liq ↔ CuSm + Sm(rhomb)	828	0.718	Eutectic
Sm(bcc) ↔ liq + Sm(rhomb)	1190	0.908	Catatectic

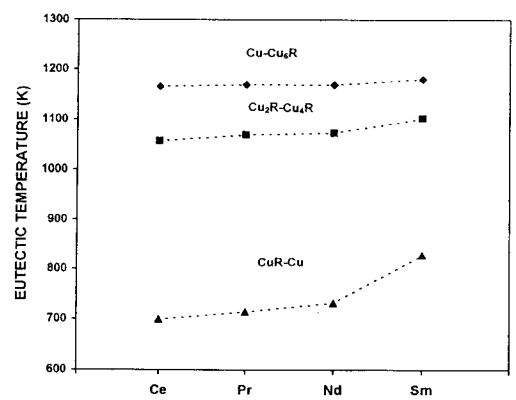

Fig. 15 Eutectic temperatures in the Cu-R systems.

Acknowledgments

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Table 10 Enthalpies of Formation of the Cu-*R* Intermediate Phases

	$\Delta_f H$, kJ/mol														
	Ce				Pr				Nd				Sm		
Phase	This work	[88Sub3]	Est.	[94Fit]	This work	[88Sub4]	Est.	[94Fit]	This work	[88Sub5]	Est.	[94Fit]	This work	[88Sub2]	Est.
Cu_6R	−10.1	−20.3	−16.0	−10.1	−12.7	−33.6	−16.1	−11.3	−13.3	−25.4	−16.0	−13.3	−13.6	−28.1	−16.3
Cu_5R	−11.5	−20.5	−18.6	...	−13.8	−30.7	−18.8	...	−14.4	−20.8	−18.6	...	−15.0	−43.2	−18.9
Cu_4R	−12.4	−16.4	−21.9	...	−15.1	−27.8	−22.2	...	−15.7	−18.1	−22.0	...	−16.6	−31.5	−22.4
Cu_2R	−14.8	−36.0	−30.9	−14.8	−19.7	−43.6	−31.5	−14.5	−20.5	−39.1	−31.4	−16.9	−21.1	−29.7	−32.0
CuR	−12.7	−32.6	−30.1	...	−19.0	−46.7	−31.0	...	−20.0	−32.1	−31.0	...	−20.9	−39.3	−31.8

Note: Mol refers to the atom as the elementary entity.

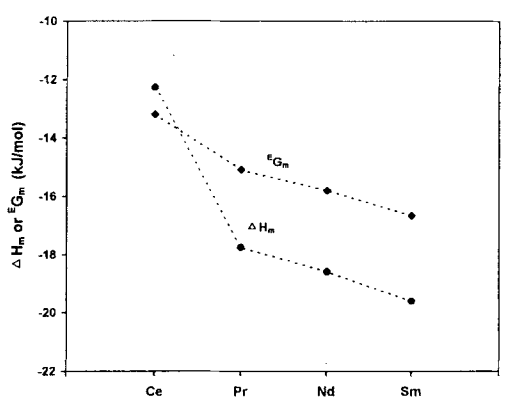

Fig. 16 Minimal values of enthalpies of mixing and excess Gibbs energies of the liquid phases in the Cu-*R* systems at 1500 K.

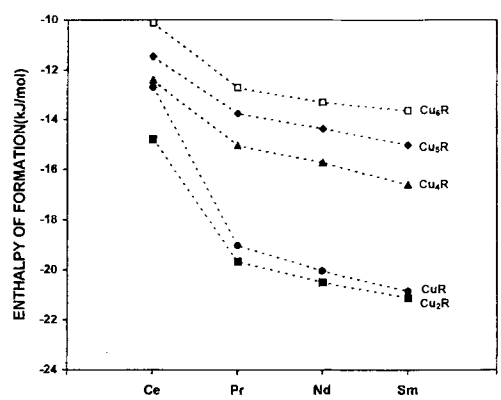

Fig. 17 Enthalpies of formation of the Cu-*R* intermediate phases.

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